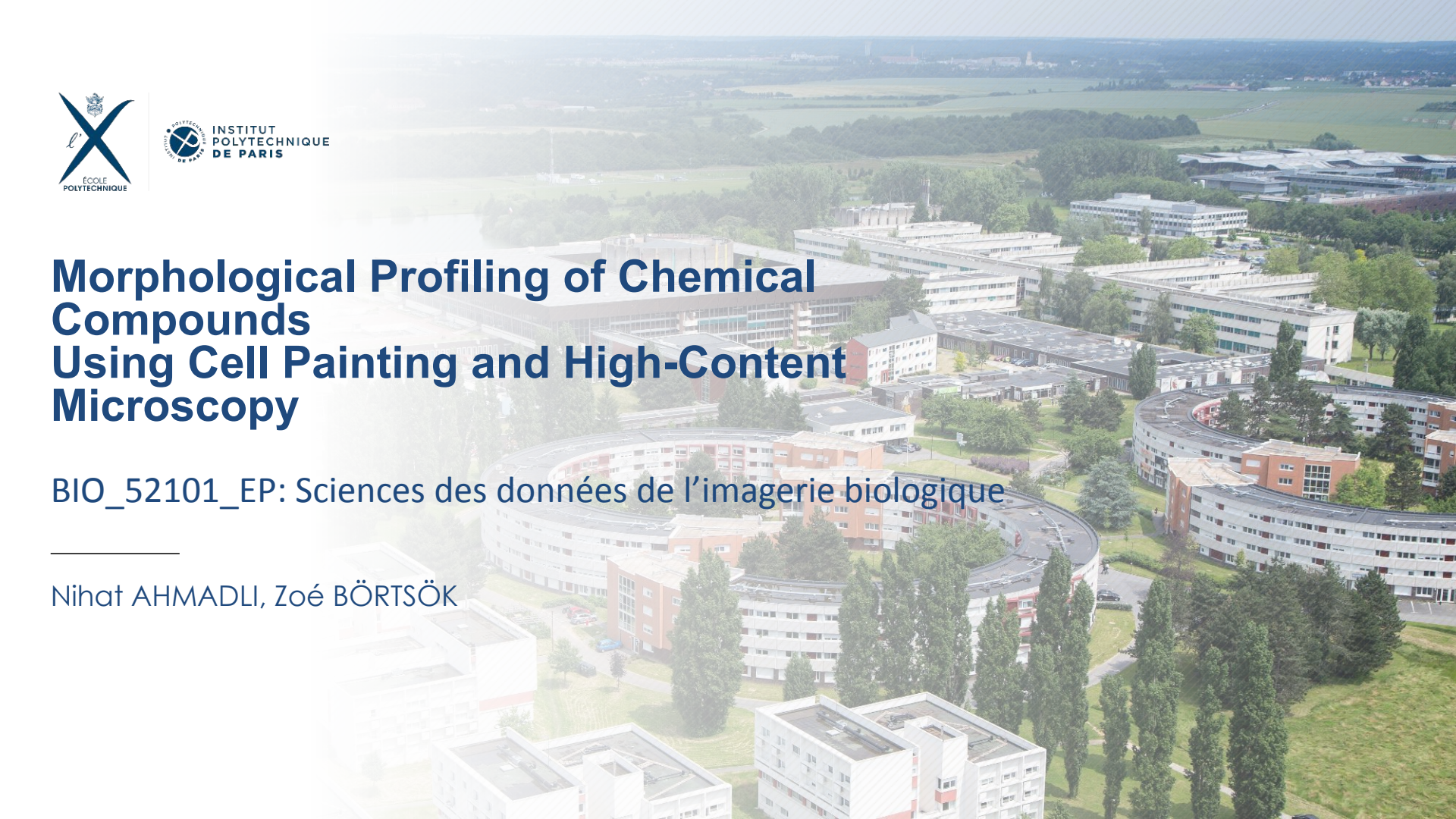


Morphological Profiling of Chemical Compounds Using Cell Painting and High-Content Microscopy

BIO_52101_EP: Sciences des données de l'imagerie biologique

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Introduction & Objectives

- **Approach:** Utilizing high-content fluorescence microscopy to conduct a data-driven analysis
- **Primary Objective:** Quantitatively assess how drug exposure alters cellular/nuclear morphology
- **Focus:** Emphasizing the biological interpretation of image-derived features and results
- **Goal:** Identify which morphological descriptors are most affected by chemical perturbations

Dataset

- **Source:** BBBC021 dataset from the Broad Bioimage Benchmark Collection (Broad Bioimage Benchmark Collection, n.d.). MCF-7 human breast cancer cells
- **Experimental Design:** Cells treated for 24 hours with 113 different compounds across 8 concentrations
- **Imaging Protocol:** Cell Painting protocol using multiplexed fluorescence imaging (DAPI channel)

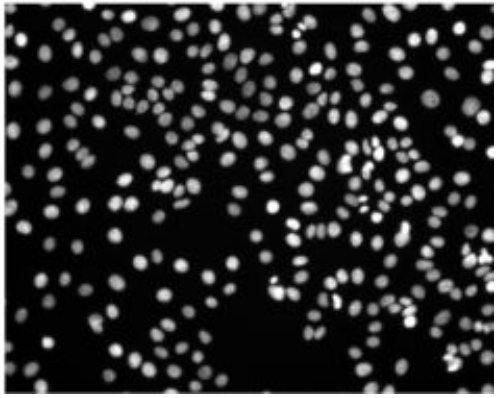


Figure 1. DMSO-treated negative-control DAPI stained fluorescent image of breast cancer cells

mageNumber	Image_FileName_DAPI	Image_PathName_DAPI	Image_FileName_Tubulin	Image_PathName_Tubulin	Image_FileName_Actin	Image_PathName_Actin	Image_Metadata_Plate_DAPI	Image_Metadata_Well_DAPI	Replicate	Image_Metadata_Compound	Image_Metadata_Concentration
233	G10_s1_w1BEDC2073-A9	Week4/Week4_27481	G10_s1_w2DCEC82F3-05F	Week4/Week4_27481	G10_s1_w43CD51CBC-23	Week4/Week4_27481	Week4_27481	G10	1	5-fluorouracil	0.003
234	G10_s2_w11C3B9BCC-E4	Week4/Week4_27481	G10_s2_w2570437EF-C8D	Week4/Week4_27481	G10_s2_w400B21F33-BI	Week4/Week4_27481	Week4_27481	G10	1	5-fluorouracil	0.003
235	G10_s3_w1F4FCE330-C7	Week4/Week4_27481	G10_s3_w2194A9AC7-369	Week4/Week4_27481	G10_s3_w4E0452054-9F	Week4/Week4_27481	Week4_27481	G10	1	5-fluorouracil	0.003

Week1_150607_C10_s1_w171173D63-FDAC-457A-9E33-9A97B4B33AB3.tif

Plate ID

Well ID & Site #

channel

Image ID

Selected Compounds & Mechanisms

- **Focus:** Comparing **DNA replication inhibitors** to a DMSO control group

- **Aphidicolin** → Replication Stress:
  - Reversibly inhibits DNA polymerase enzymes
  - Blocks S phase of the cell cycle
  - Stalls replication forks and generates single-stranded DNA (Krokan et al., 1981)

- **Mitomycin C** (MMC) → DNA Cross-linking:
  - Highly reactive DNA cross-linking agent
  - Forms intrastrand cross-links, blocking replication and transcription (Verweij & Pinedo, 1990)

chosen 2 drugs

! Chosen **concentrations** for each compound: **0.003** & **10 µM**

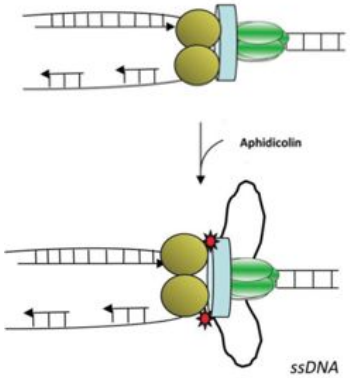


Figure 2. Aphidicolin MoA (Recolin et al., 2014)

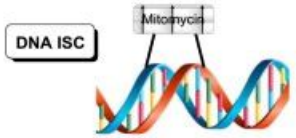


Figure 3. MMC MoA (Demirdağ et al., 2012)

Expected Effects on Nuclear Morphology

- **Aphidicolin Effects:**
  - Nuclear swelling (chromatin evenly distributed)
  - Prolonged stress can lead to DNA double-strand breaks (Vesela et al., 2017)
- **Mitomycin C Effects:**
  - High genotoxicity triggers programmed cell death (apoptosis)
  - Leads to nuclear fragmentation, membrane blebbing, and chromatin condensation (Ma, 2005)

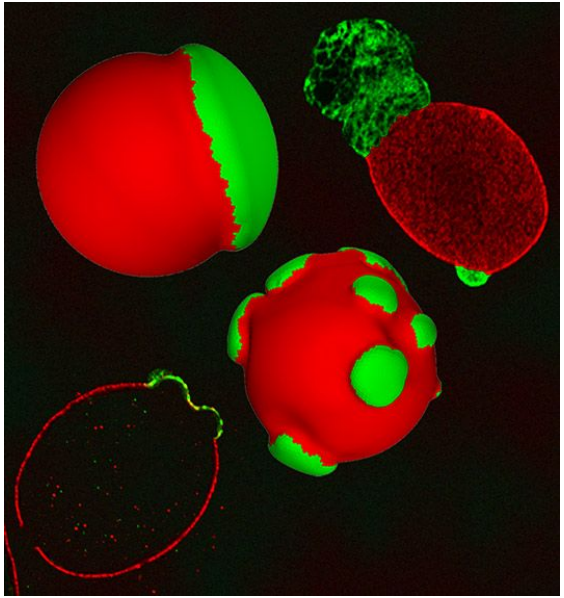


Figure 4. Simulation of Nuclear Blebbing
(Funkhouser et al., 2013)

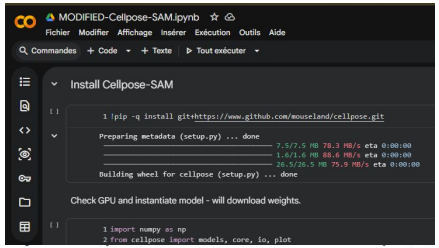
Methodology & Analytical Pipeline

- **Extraction of relevant data and image**
  - Negative controls (DMSO)
  - Aphidicolin (0.003 & 10 µL)
  - MMC (0.003 & 10 µL)

  ❗ 3 replicates/plates for each drug concentration

- **Segmentation of nucleus** using Cellpose-SAM

- **Analytical Steps:**
  1. **Feature extraction** regarding the cell morphology
  2. **Comparative visualizations** (violin/box plots), PCA, and statistical tests
  3. Confirm the drug effect on the cell morphology and present the results



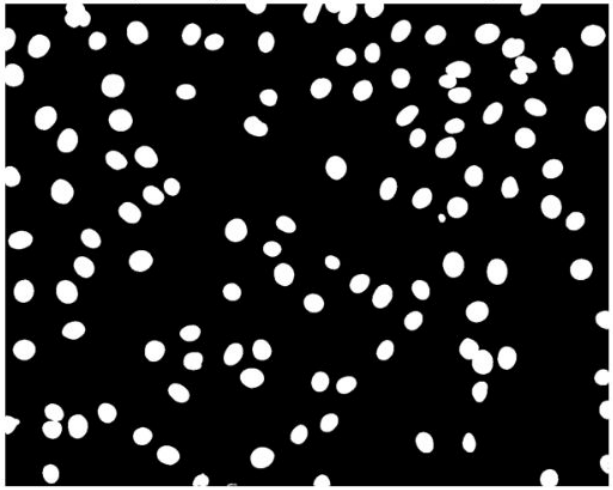


Figure 5. Cellulose-SAM Generated Binary Mask (Aphidocolin C=10µL)

Feature Extraction

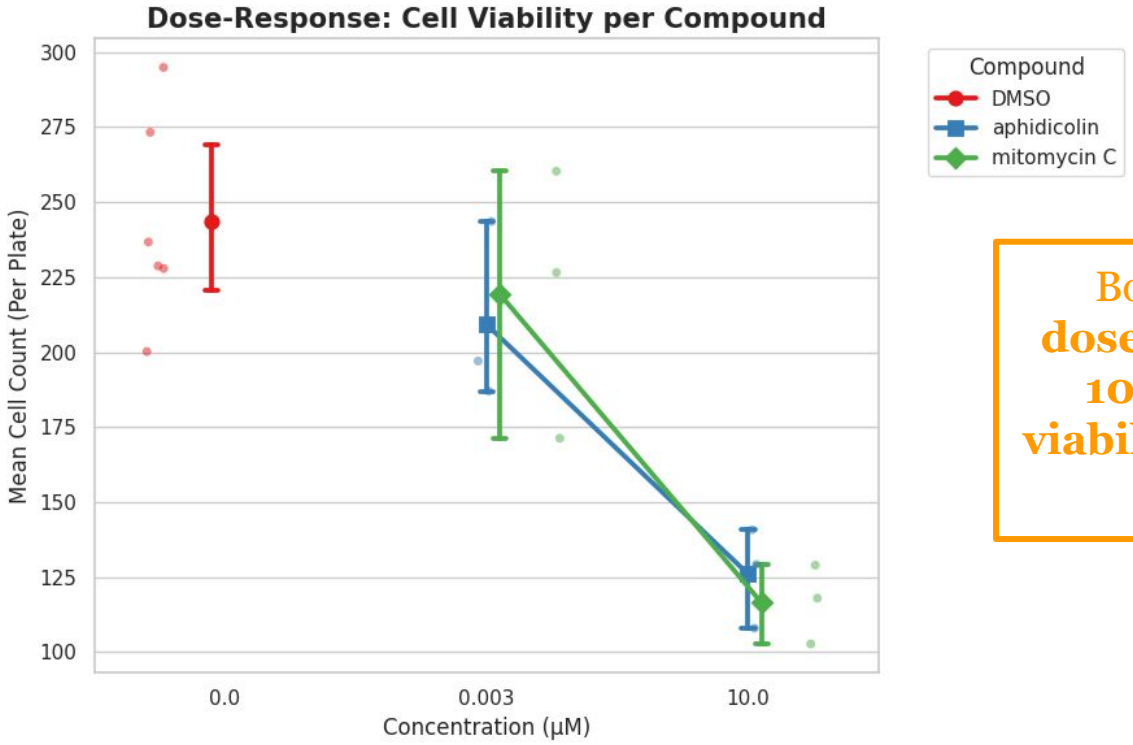
- **Step 1:** Single-Cell Feature Extraction
  - Extracted 20+ morphological and intensity metrics per cell (e.g: Area, circularity, mean intensity, aspect ratio, etc.)
- **Step2:** Apply across wells to get values per well
  - Loop through and extract everything into 1 df
  - Add metadata to the well_profiles

	area	perimeter	eccentricity	solidity	extent	major_axis_length	minor_axis_length	circularity	aspect_ratio	compactness	...	area_variance	well_name	cell_count	compound	concentration	Plate ID
0	947.026667	112.901830	0.624406	0.971306	0.762951	39.339381	29.853260	0.895180	1.413592	14.211562	...	94696.222500	Week1_150607_B02_s1_w107447158-AC76-4844-8431-...	225	DMSO	0.0	Week1_22123
1	850.820652	107.114287	0.653380	0.965275	0.754027	37.738855	27.649595	0.873999	1.487027	14.696511	...	116757.841976	Week1_150607_B02_s2_w14AB22A28-7EF5-461A-8956-...	184	DMSO	0.0	Week1_22123
2	984.158273	115.718500	0.615590	0.971268	0.767361	40.257938	30.577399	0.892152	1.396976	14.239326	...	96780.227567	Week1_150607_B02_s3_w1655B02F9-3E1A-44E4-ADA1-...	278	DMSO	0.0	Week1_22123
3	891.943709	110.113336	0.621961	0.971281	0.762902	38.352658	29.072559	0.888728	1.408267	14.318829	...	74921.408781	Week1_150607_B02_s4_w1EB868A72-19FC-46BB-BDFA-...	302	DMSO	0.0	Week1_22123
4	939.089362	112.536085	0.617664	0.970736	0.765344	39.113503	29.816698	0.892651	1.388049	14.246565	...	92068.534715	Week1_150607_C02_s1_w13355670E-7C0E-404D-9EB5-...	235	DMSO	0.0	Week1_22123

5 rows × 29 columns

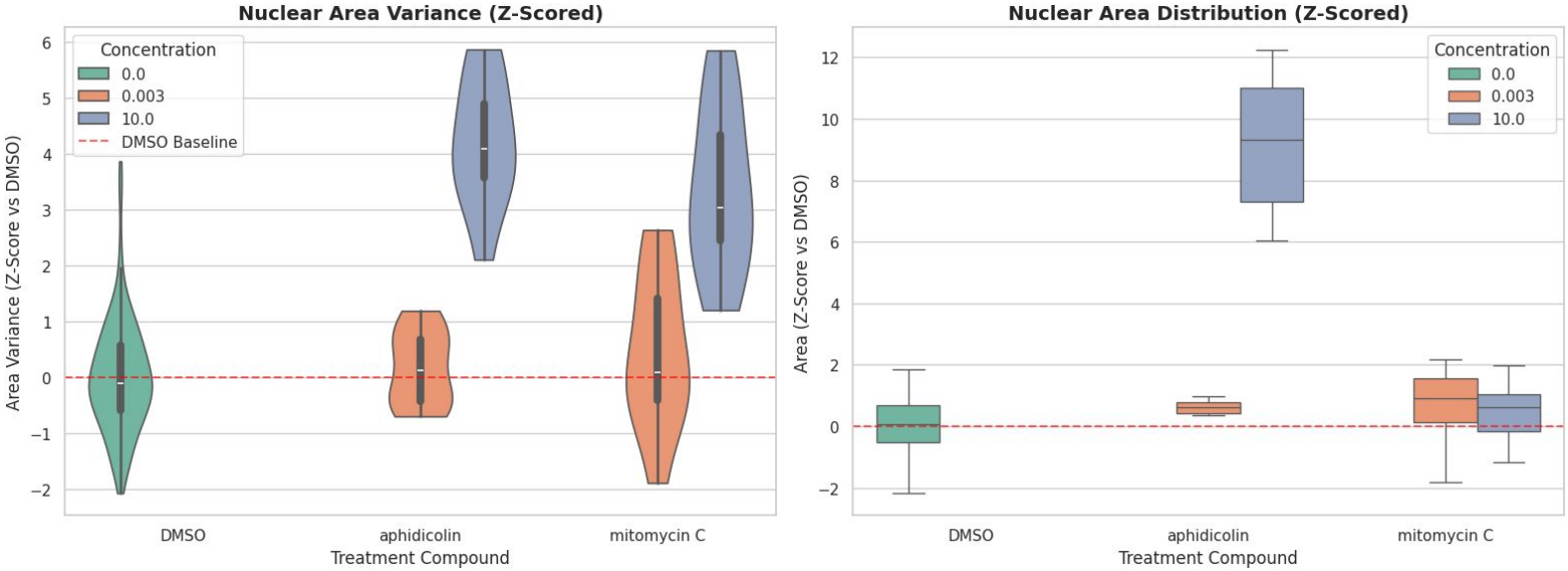
- **Step 3:** Batch Normalization w.r.t plates
  - Aggregated data to the well/image level
  - Calculated Z-scores per plate strictly against the internal DMSO negative control
  - Thus, mathematically eliminated plate-to-plate variance (plate effect)

Results - Cell Viability



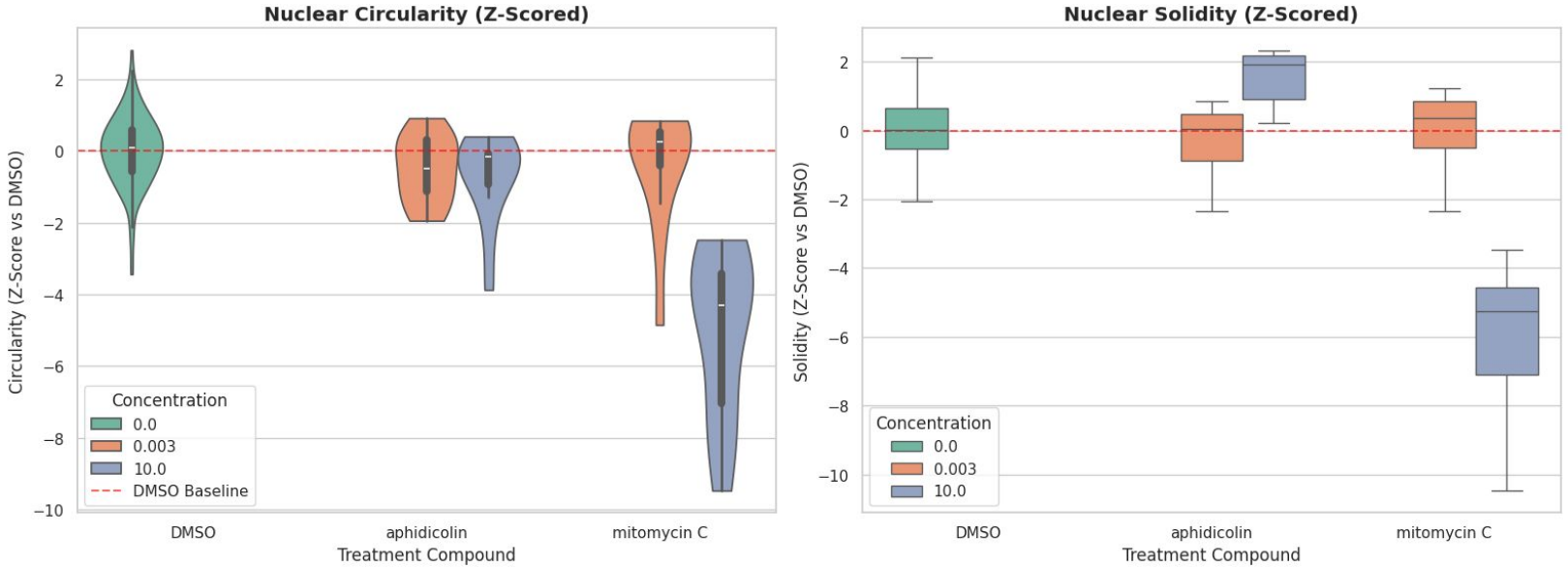
Both compounds show **dose-dependent toxicity, 10.0 µM reduces cell viability by ~50%** compared to DMSO

Results - Nuclear Size



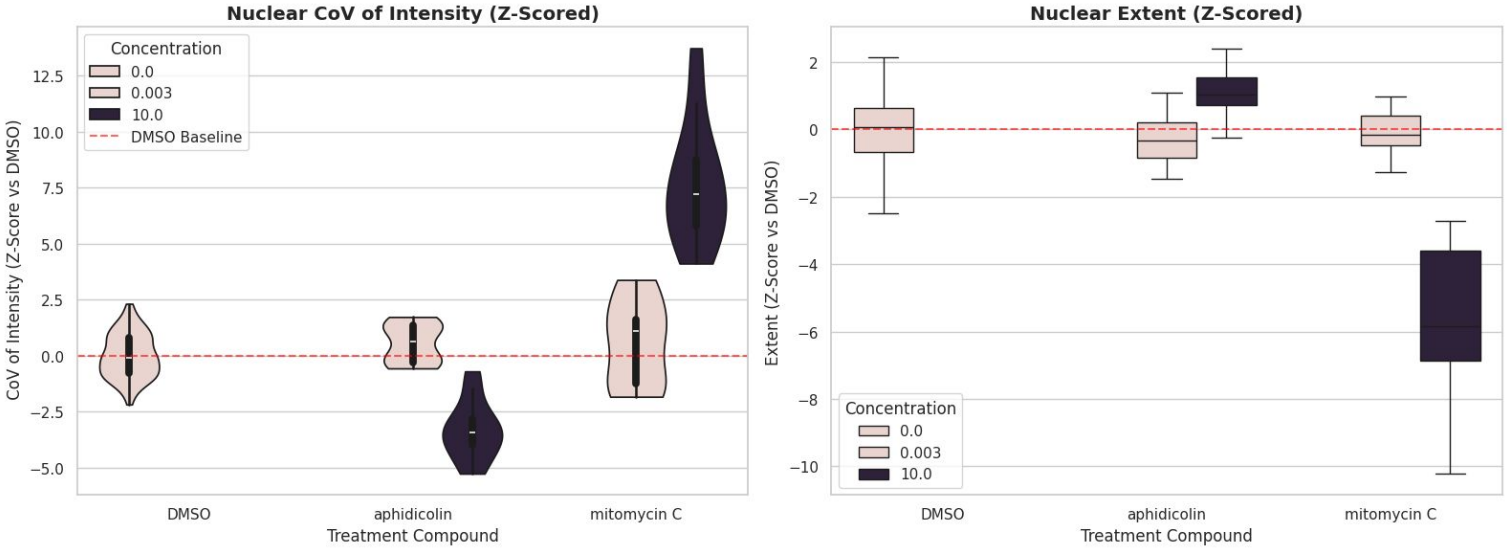
High-dose **aphidicolin causes nuclear swelling**, whereas **MMC increases area variance** and showing little change in area

Results - Shape & Texture Alterations



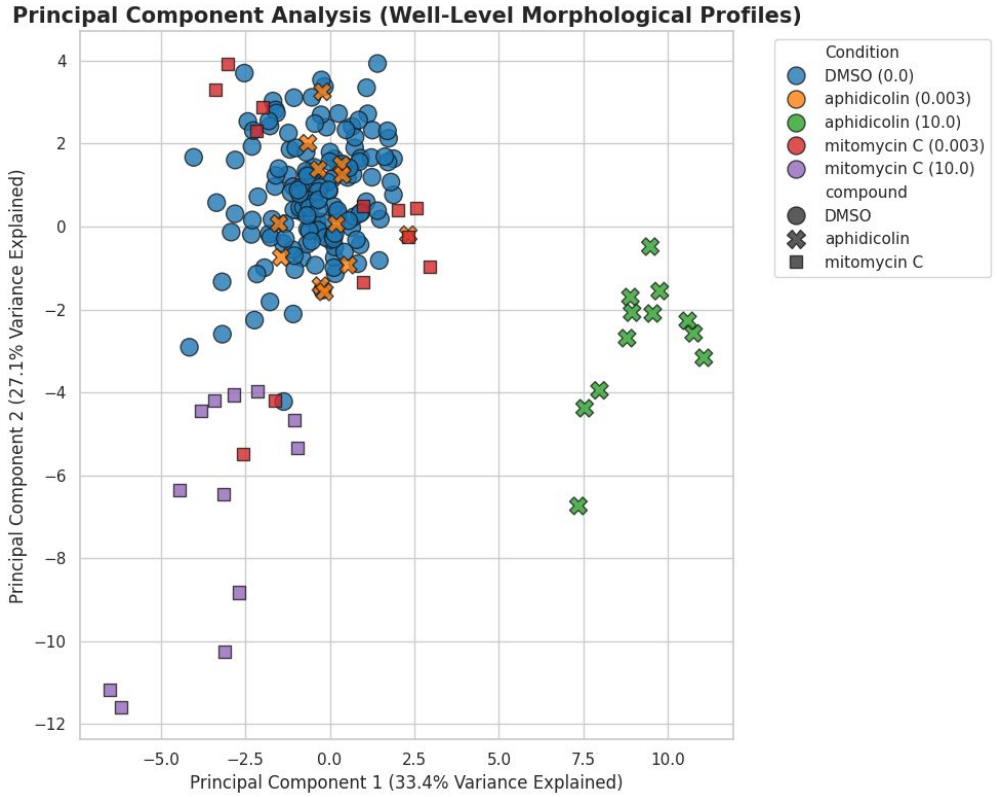
High-dose MMC results in a **loss of structural integrity**, marked by a drop in circularity and solidity, probably due to nuclear blebbing and fragmentation.

Results - DAPI Intensity and Nuclear Extent Change



The high intensity variation and shrunken extent of **MMC** suggest **apoptotic DNA clumping (chromatin condensation)**, while the shrunken intensity and increased extent of **Aphidicolin**-treated cells highlight **smooth nuclear expansion (chromatin evenly distributed across nucleus)**.

Results - Principal Component Analysis (PCA)



From PCA, we observe distinct clustering of our drugs. This further showcases high-dose treatments with Aph and MCM induce distinct morphological profiles:

- **Aphidicolin** drives changes characterized by **nuclear swelling** along PC1,
- **MMC drives texture and shape irregularities** along PC2.

Results - Statistical Validation - ANOVA

- We apply **ANOVA** on **mean nuclear area** and **area variance** to compare 3 groups
- We also apply **Tukey HSD** to examine which groups differ if ANOVA gives a **significant result**

```
--- BALANCED OBSERVATIONS (Images) PER GROUP ---
Group
DMSO (0.0)           12
aphidicolin (10.0)   12
mitomycin C (10.0)   12
Name: count, dtype: int64


--- ONE-WAY ANOVA ON AREA ---
F-statistic: 141.38
P-value: 6.55e-17

--- TUKEY HSD POST-HOC TEST ON AREA ---
           Multiple Comparison of Means - Tukey HSD, FWER=0.05
=================================================================================
      group1            group2       meandiff p-adj   lower    upper  reject
---------------------------------------------------------------------------------
       DMSO (0.0) aphidicolin (10.0)   9.1605    0.0   7.6447 10.6762   True
       DMSO (0.0) mitomycin C (10.0)   0.3393 0.8475  -1.1764  1.8551  False
aphidicolin (10.0) mitomycin C (10.0)  -8.8212    0.0 -10.3369 -7.3054   True
---------------------------------------------------------------------------------
```

```
--- BALANCED OBSERVATIONS (Images) PER GROUP ---
Group
DMSO (0.0)           12
aphidicolin (10.0)   12
mitomycin C (10.0)   12
Name: count, dtype: int64


--- ONE-WAY ANOVA ON AREA Variance ---
F-statistic: 42.16
P-value: 8.15e-10

--- TUKEY HSD POST-HOC TEST ON AREA Variance ---
           Multiple Comparison of Means - Tukey HSD, FWER=0.05
=================================================================================
      group1            group2       meandiff p-adj  lower  upper  reject
---------------------------------------------------------------------------------
       DMSO (0.0) aphidicolin (10.0)   4.2021   0.0  3.0157 5.3884   True
       DMSO (0.0) mitomycin C (10.0)   3.3416   0.0  2.1552 4.5279   True
aphidicolin (10.0) mitomycin C (10.0)  -0.8605 0.192 -2.0469 0.3258  False
---------------------------------------------------------------------------------
```

Statistical Validation - ANOVA

```
--- BALANCED OBSERVATIONS (Images) PER GROUP ---
Group
DMSO (0.0)            12
aphidicolin (10.0)    12
mitomycin C (10.0)    12
Name: count, dtype: int64


--- ONE-WAY ANOVA on Circularity ---
F-statistic: 36.27
P-value: 4.67e-09

--- TUKEY HSD POST-HOC TEST ON Circularity ---
           Multiple Comparison of Means - Tukey HSD, FWER=0.05
=================================================================================
      group1            group2       meandiff p-adj  lower   upper  reject
---------------------------------------------------------------------------------
       DMSO (0.0) aphidicolin (10.0)  -0.7145 0.503 -2.2665  0.8375  False
       DMSO (0.0) mitomycin C (10.0)  -4.9809   0.0 -6.5329 -3.4289   True
aphidicolin (10.0) mitomycin C (10.0)  -4.2664   0.0 -5.8184 -2.7144   True
---------------------------------------------------------------------------------
```

```
--- BALANCED OBSERVATIONS (Images) PER GROUP ---
Group
DMSO (0.0)            12
aphidicolin (10.0)    12
mitomycin C (10.0)    12
Name: count, dtype: int64


--- ONE-WAY ANOVA on Solidity ---
F-statistic: 82.79
P-value: 1.38e-13

--- TUKEY HSD POST-HOC TEST ON Solidity ---
           Multiple Comparison of Means - Tukey HSD, FWER=0.05
=================================================================================
      group1            group2       meandiff p-adj  lower   upper  reject
---------------------------------------------------------------------------------
       DMSO (0.0) aphidicolin (10.0)   1.3783 0.076 -0.1186  2.8752  False
       DMSO (0.0) mitomycin C (10.0)  -6.0031   0.0    -7.5 -4.5063   True
aphidicolin (10.0) mitomycin C (10.0)  -7.3814   0.0 -8.8783 -5.8846   True
---------------------------------------------------------------------------------
```

Conclusion & Biological Interpretations

- Successfully quantified morphological differences distinguishing DNA damage agents like aphidicolin and MMC from controls

- **Biological Link:**
  - **Aphidicolin** → nuclear swelling (size increases)
  - **MMC structurally degrades the nucleus shape** → lowering circularity and solidity

- High-content imaging successfully captures specific mechanistic actions of drugs. Both compounds are dose-dependent inhibitors of cell growth, but they act through different mechanisms, proved by our results

- Our results matches the Literature Findings



Thank you for your attention!

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